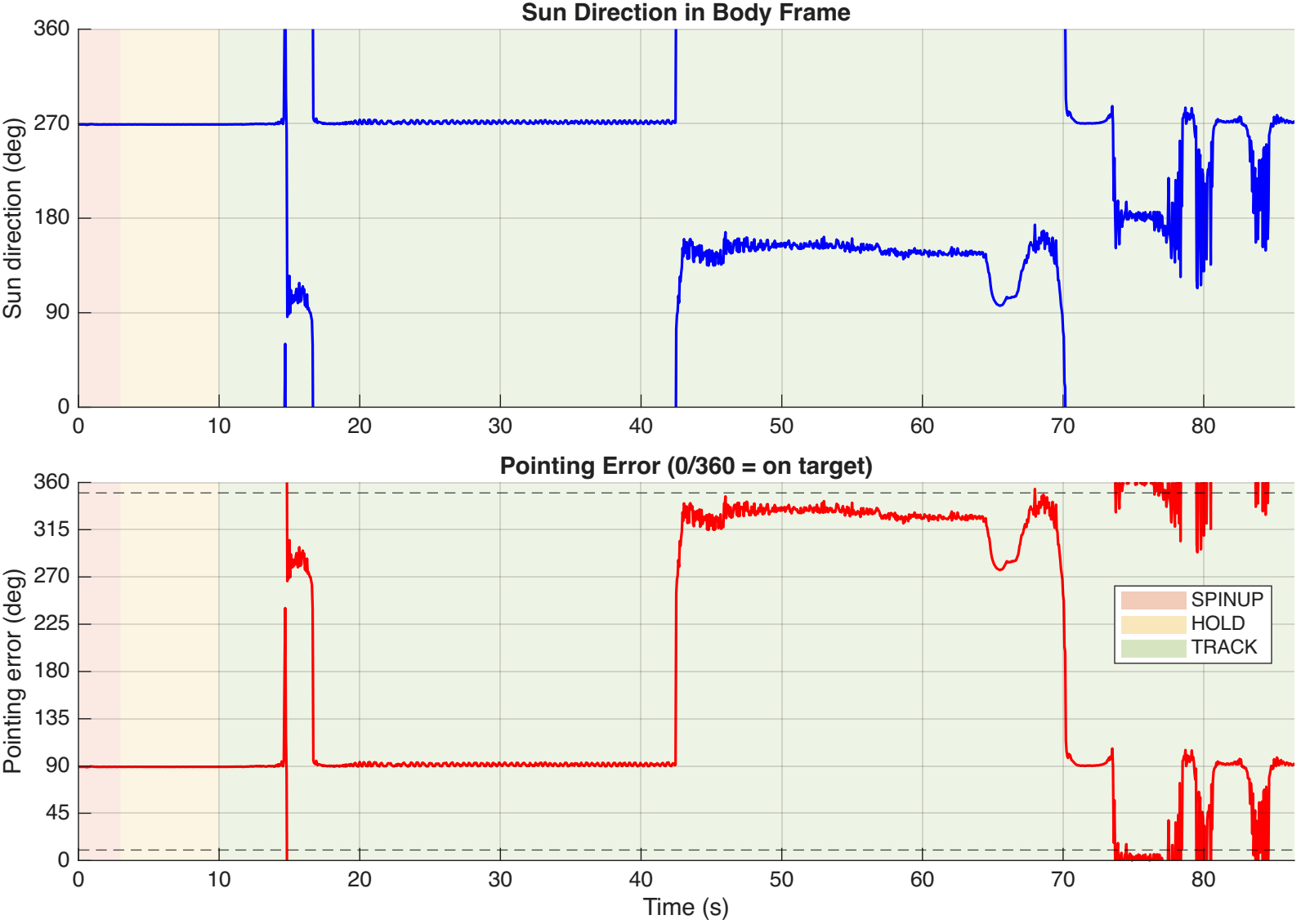
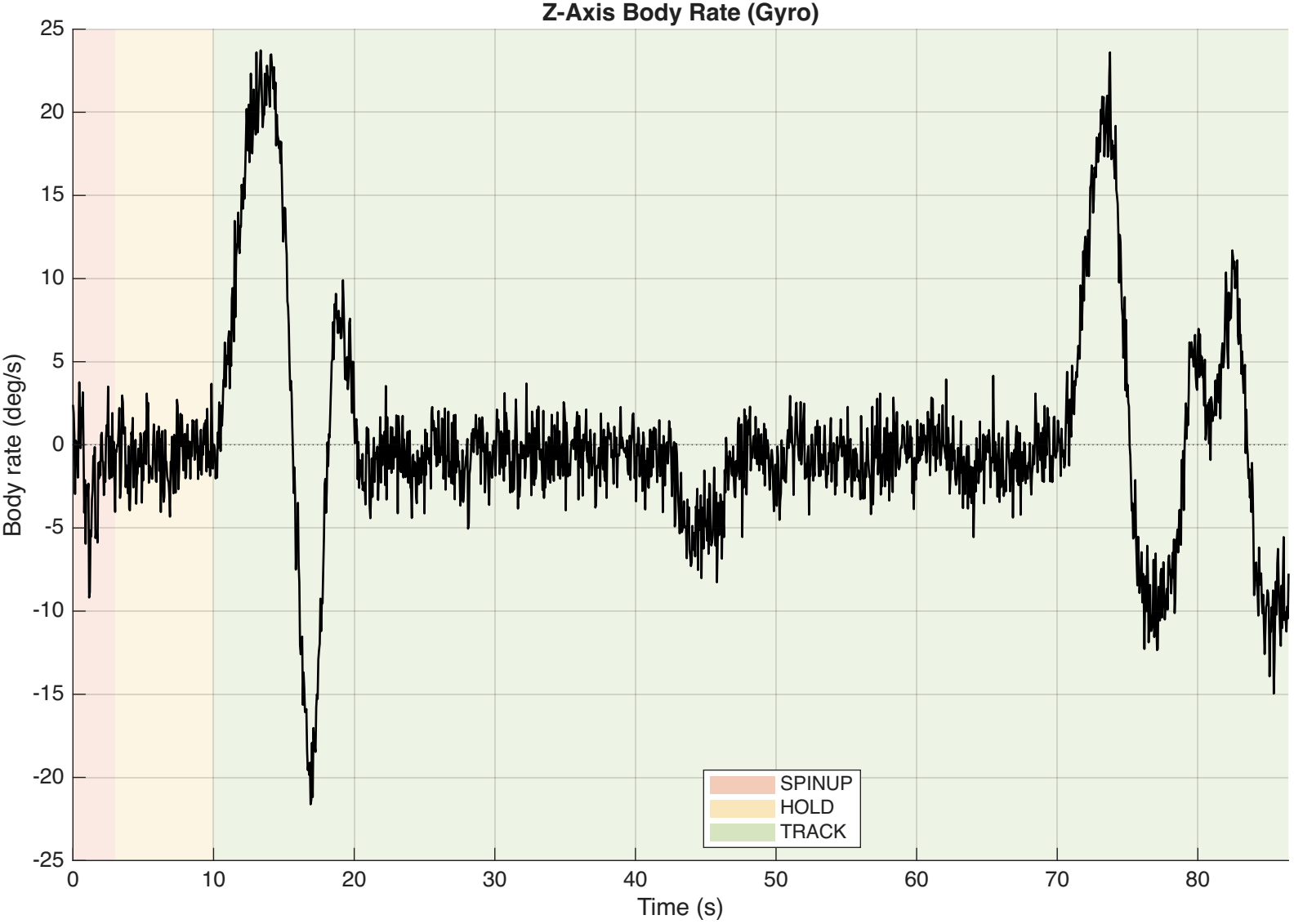
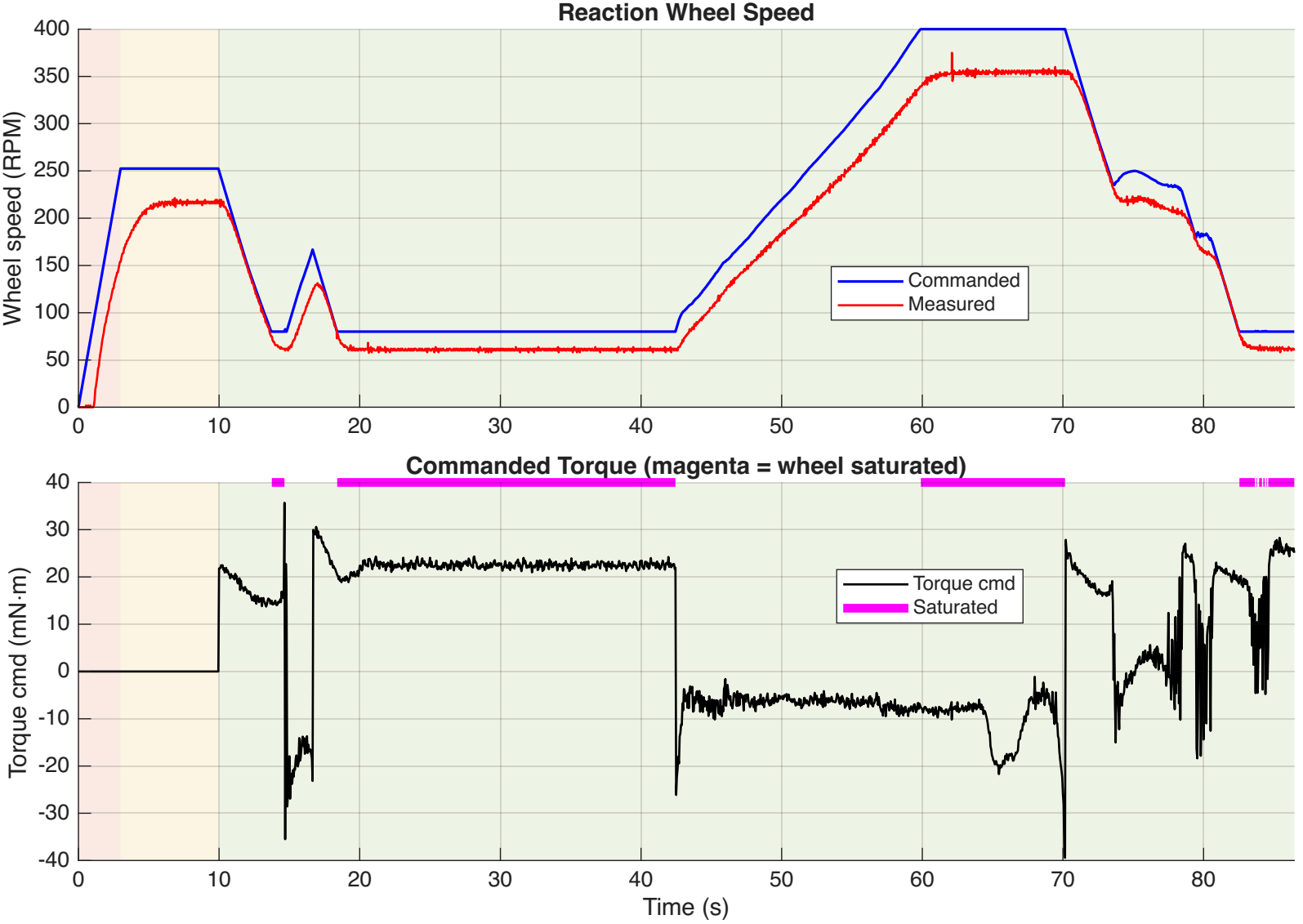
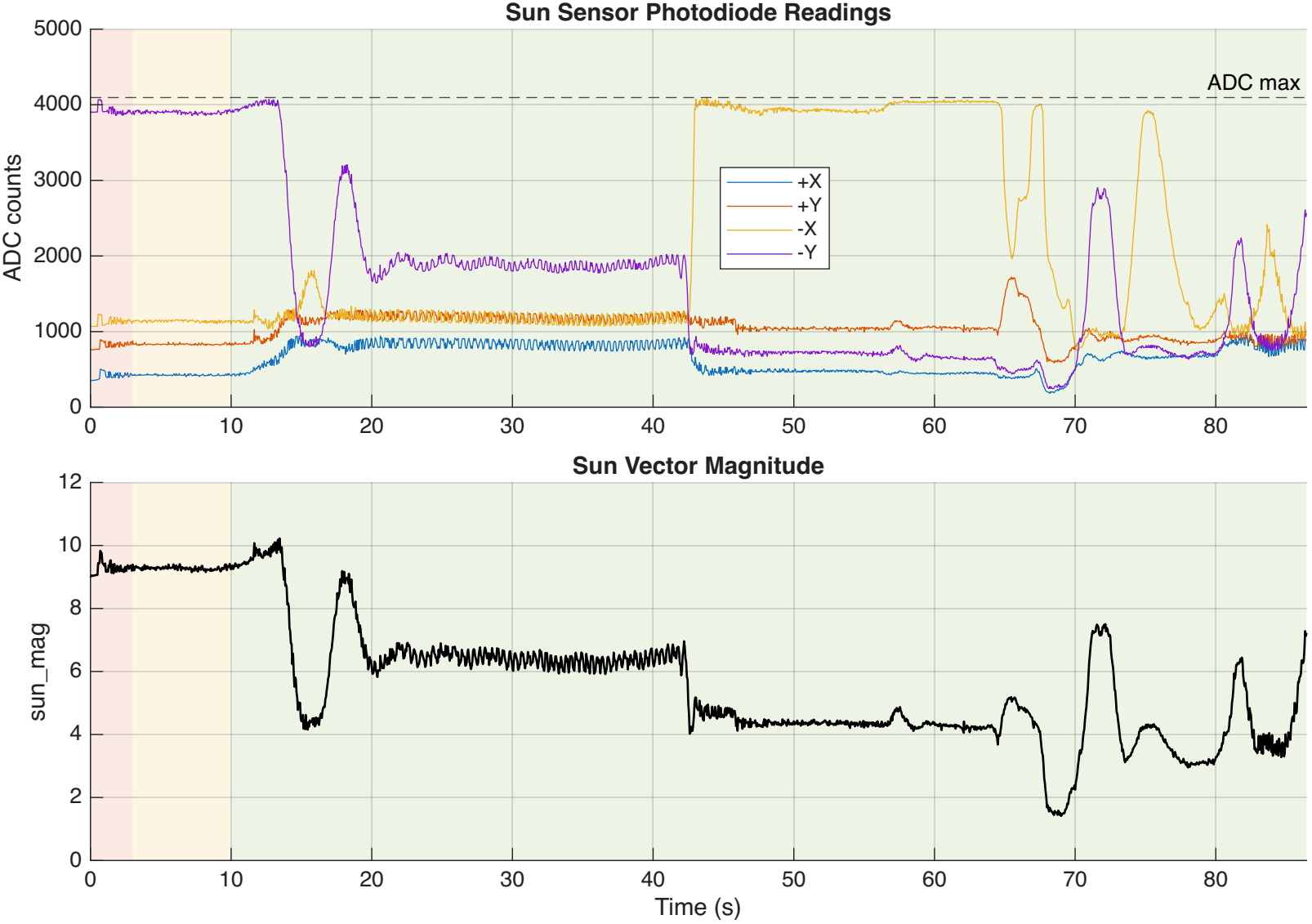
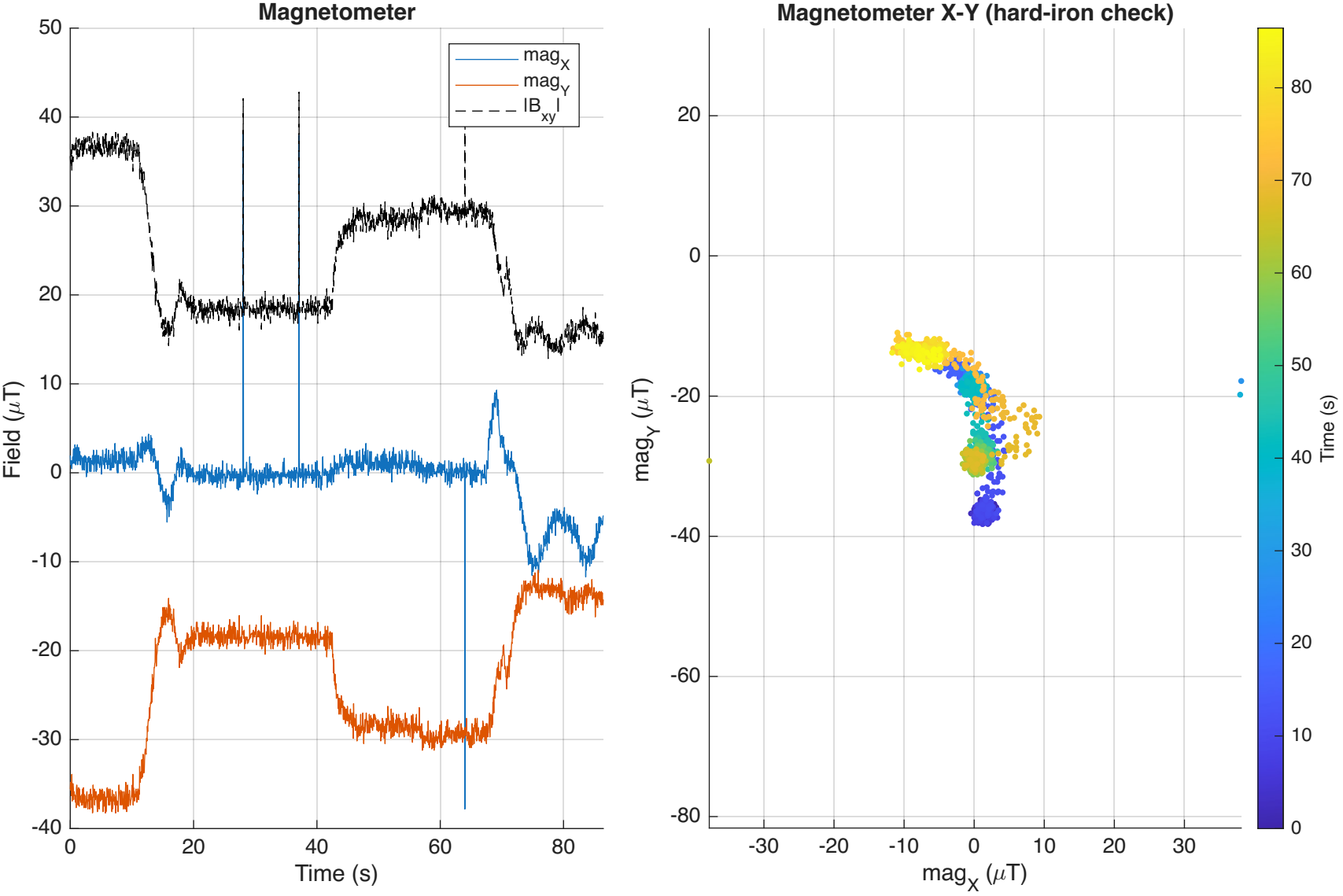
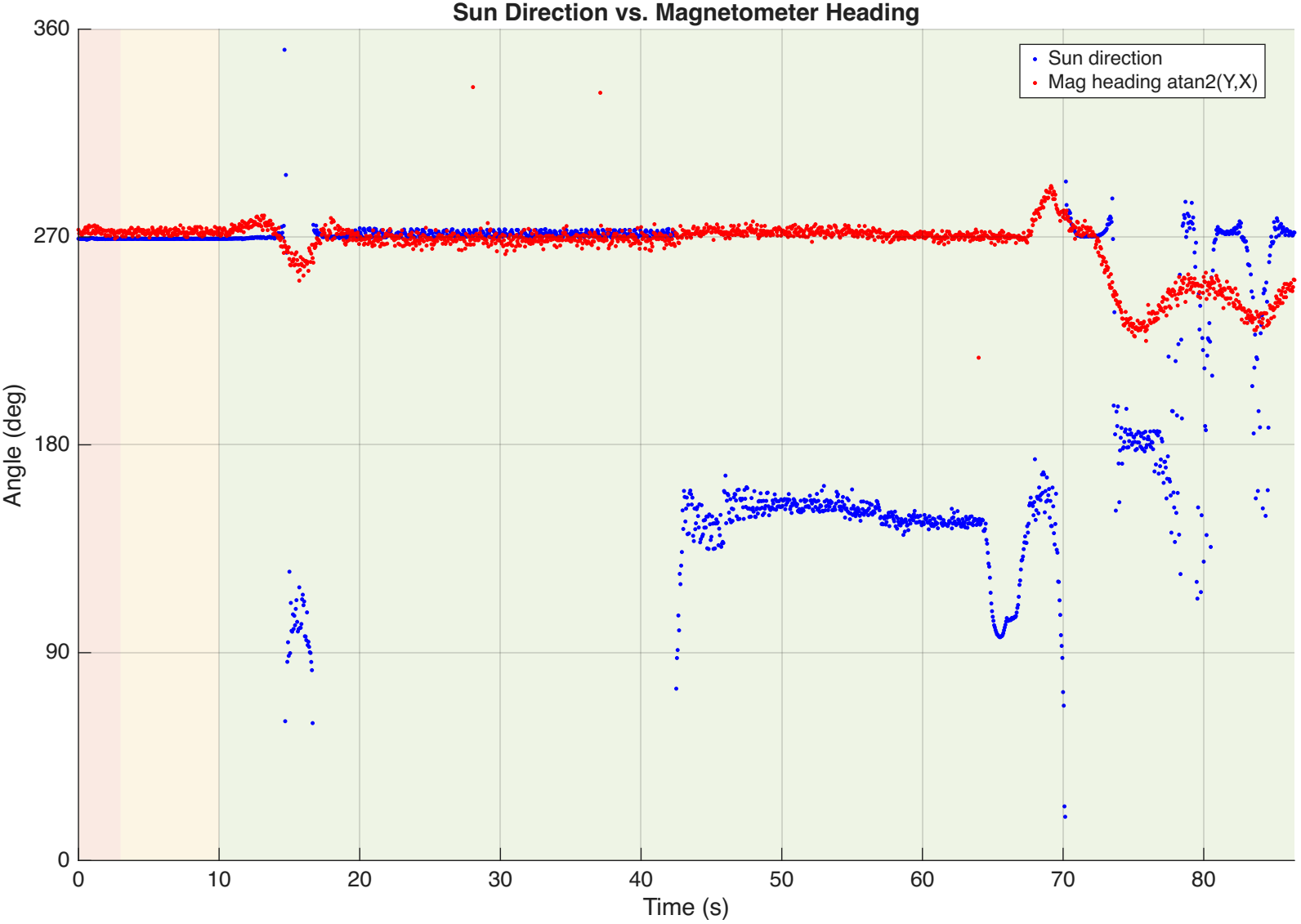

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
10
20
30
40
50
60
70
80
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)


Z-Axis Body Rate (Gyro)
Body rate (deg/s)
Time (s)
25
20
15
10
5
0
-5
-10
-15
-20
-25
0
10
20
30
40
50
60
70
80
SPINUP
HOLD
TRACK


Reaction Wheel Speed
Wheel speed (RPM)
400
350
300
250
200
150
100
50
0
0
10
20
30
40
50
60
70
80
Commanded
Measured
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
40
30
20
10
0
-10
-20
-30
-40
Torque cmd
Saturated
Time (s)


Sun Sensor Photodiode Readings
ADC counts
ADC max
+X
+Y
-X
-Y
Sun Vector Magnitude
sun_mag
Time (s)


Magnetometer
mag_X
mag_Y
|B_xy|
Field (μT)
Time (s)
Magnetometer X-Y (hard-iron check)
mag_Y (μT)
mag_X (μT)
Time (s)


Sun Direction vs. Magnetometer Heading
Sun direction
Mag heading atan2(Y,X)
Angle (deg)
360
270
180
90
0
0
10
20
30
40
50
60
70
80
Time (s)


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
-X face (solar array)
Sun (from sun sensor)
0
10
20
30
40
50
60
70
80
KestrelSAT Rotation (integrated gyro)
ψ (deg)
70
60
50
40
30
20
10
0
-10
Time (s)